

Tailored and Tunable Microwave Wireless Power Transfer for Multiple Targets in an Electrically Large Cavity via Frequency Control

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Abstract—This study introduces a novel approach to microwave wireless power transfer (MWPT) that enables precise and customizable charging for multiple targets within an electrically large cavity (ELC). By adjusting frequency, the electromagnetic (EM) field distribution within the cavity is dynamically controlled, facilitating targeted power delivery to specific locations. Simulation results demonstrate that the EM field exhibits high sensitivity to frequency variations, leading to significant changes in field strength at various target positions. The capability to simultaneously adjust both frequency and input power enables adaptive and efficient wireless charging, precisely matching the energy requirements of targets. To validate the proposed method, an experiment setup comprising 160 light-emitting diode (LED) rectennas was arranged inside the ELC, enabling tailored MWPT for multiple targets. The experimental results closely align with simulations. As a proof of concept, LED rectennas arranged along a specific path were sequentially illuminated, validating the effectiveness of the frequency-control scheme in achieving customized multitarget MWPT.

Index Terms—Electrically large cavity, microwave wireless power transfer (MWPT), multitargets, rectenna, tailored wireless charging.

I. INTRODUCTION

MICROWAVE wireless power transfer (MWPT) technology offers notable advantages, such as long-distance

energy delivery and targeted charging capabilities, making it highly suitable for applications including medical devices, smart home systems, wireless sensor networks (WSNs), and the Internet of Things (IoT) [1], [2], [3], [4], [5], [6], [7]. Significant progress has been achieved in MWPT in free space, covering key areas such as transmission, receiving arrays, and rectification [8], [9], [10], [11], [12], [13], [14], [15].

In applications such as aircraft, vehicles, rockets, and industrial facilities, the widespread deployment of IoT often requires numerous sensors [16], [17]. However, ensuring continuous power delivery to these sensors presents major challenges, particularly in constrained or enclosed environments where wired power solutions are impractical and battery replacements are costly. Addressing these issues, MWPT technology, particularly within electrically large cavities (ELCs), has emerged as a promising solution, primarily due to significantly reduced radiation losses [19], [20], [21], [22], [23], [24].

Several studies have demonstrated the effectiveness of WPT within these enclosures [25], [26], [27], [28], [29], [30], [31], [32], [33], [34]. Common methods include quasi-static cavity resonance (QSCR) [25], multimode method [27], [30], and probes with variable reactance [22], [29], [31]. For example, in [27], MWPT at 0.45 GHz was implemented within a multimode resonant enclosure with a volume of $0.6\lambda^3$, achieving MWPT for four targets. Another study employed reflection probes in a rectangular cavity ($4.36\lambda^3$ at 2.45 GHz) to improve transfer efficiency at blind spots [31].

Previous studies have predominantly focused on MWPT experiments conducted within resonant enclosures smaller than $5\lambda^3$, where electromagnetic (EM) fields typically exhibit standing-wave distributions. However, in larger-scale environments like the workplaces of reusable rockets, the volume is approximately 4.5 m^3 [17]. MWPT in ELCs commonly operates at frequencies such as 2.45 GHz, 5.8 GHz, or even higher. Within these cavities, the EM field distributions become significantly complex and highly sensitive to frequency and geometry [32]. Therefore, effectively utilizing the field distribution characteristics within ELCs is crucial for achieving tailored multitarget charging to meet the growing demand for wireless charging in sensor networks. Nonetheless, only a few studies have explored this topic. For instance, [33] reported in 2014 employed mechanical stirring, conductive and absorptive object loading, and

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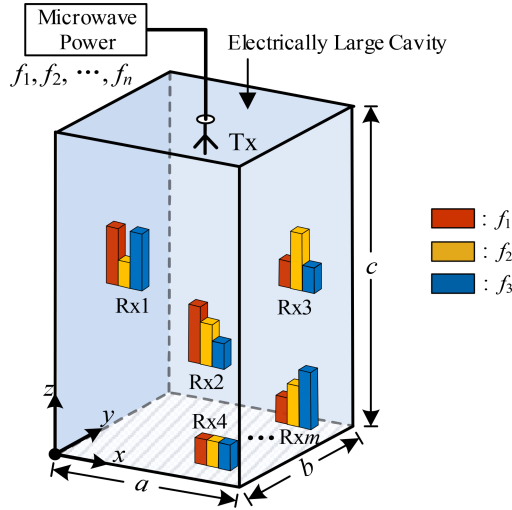


Fig. 1. Schematic diagram of frequency-controlled MWPT for multiple targets in an electrically large cavity ($a = 1.3$ m, $b = 1.3$ m, and $c = 2$ m).

frequency modulation to achieve a relatively uniform EM field distribution at 10 GHz. Moreover, these studies primarily focus on large-area wireless power delivery and rarely address tailored wireless charging for multiple targets.

This work presents a novel MWPT method that enables tailored and tunable charging for multiple targets within an ELC. The key innovation lies in dynamically tuning the frequency to reshape the EM field distribution, ensuring precise and customizable energy delivery to each target at different locations. Compared to conventional MWPT in ELCs, the proposed approach significantly enhances targeted power transfer by adapting to the unique energy demands of selected targets. Both simulations and experiments validate its feasibility.

II. MULTITARGET TUNABLE MWPT METHOD

A. Schematic Diagram of Multitarget Tunable MWPT

A schematic diagram of the multitarget tunable MWPT system is shown in Fig. 1. The rectangular ELC has dimensions of $1.3 \text{ m} \times 1.3 \text{ m} \times 2 \text{ m}$, yielding a volume of 3.38 m^3 , which corresponds to $1.83 \times 10^3 \lambda^3$ at 2.45 GHz. The relationship between the number of modes (N_W) and frequency (f) is given by [32]

$$N_W(f) = \frac{8}{3} \pi abc \left(\frac{f}{v} \right)^3 \quad (1)$$

where v represents the speed of light in vacuum ($3 \times 10^8 \text{ m/s}$), while a , b , and c denote the length, width, and height of the cavity, respectively. The typical frequency shift (Δf) between adjacent modes is expressed as

$$\Delta f = \frac{1}{D_W(f)} = \frac{1}{\frac{dN_W(f)}{df}} = \frac{v^3}{8\pi abc f^2} \quad (2)$$

where D_W is the mode density (modes/Hz), and f is 2.45 GHz. The frequency shifter Δf is approximately 53 kHz, indicating

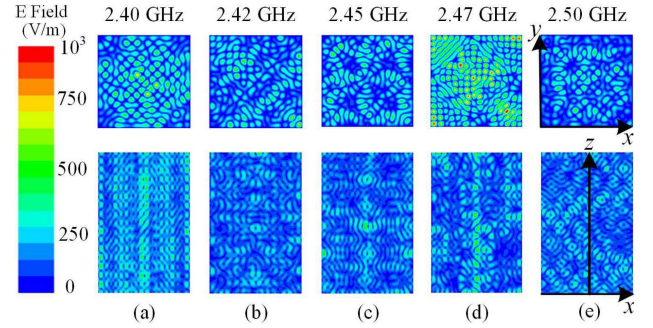


Fig. 2. Simulated E-field distributions in the ELC at (a) 2.40 GHz; (b) 2.42 GHz; (c) 2.45 GHz; (d) 2.47 GHz; and (e) 2.50 GHz.

that even a slight frequency variation can significantly change the EM field distribution within the cavity.

As illustrated in Fig. 1, a transmitting antenna is placed at the top center of the ELC to emit EM waves across a range of discrete frequencies (e.g., $f_1, f_2, \dots, f_n$). Multiple targets ($Rx_1, Rx_2, \dots, Rx_m$) are spatially distributed throughout the ELC. Color-coded bars are used to represent the incident frequencies and the corresponding electric field (E-field) intensities at each target location. For instance, red, yellow, and blue indicate frequencies f_1, f_2 , and f_3 , respectively, while the bar heights reflect relative E-field magnitudes. As an example, Rx_1 experiences the strongest E-field at f_1 and the weakest at f_3 , whereas Rx_4 receives low E-field levels at all three frequencies.

By tuning the transmitter's frequency and power P_t , selective and flexible power delivery can be achieved, either simultaneously to multiple targets or individually to a specific target. For example, at f_1 , both Rx_1 and Rx_2 experience elevated E-field strength, whereas at f_2 , only Rx_3 receives significant power. As higher E-field intensity generally implies greater received power, coordinated tuning of frequency and P_t enables precise control of power distribution. In summary, the proposed MWPT approach enables tailored power delivery to specific targets by dynamic adjustment of frequency and P_t .

B. Simulation Results and Analysis

Determining the optimal frequency for customized wireless charging of multiple targets is challenging due to the high sensitivity of EM field distribution within ELCs to frequency variation. To address this, the electric field (E-field) distributions in a perfect electric conducting (PEC) rectangular ELC at various frequencies were simulated using Ansys High-Frequency Structure Simulator (HFSS). The results in Fig. 2 show that the E-field distributions on the x - y ($z = 0$) and x - z ($y = b/2$) planes vary significantly across 2.4 GHz to 2.5 GHz. While the E-field patterns are more uniform than in typical resonant cavities, significant variations remain, which can be exploited for frequency-tuning-based tailored MWPT in ELCs. In particular, the E-field magnitude peaks at 2.47 GHz, attributed to the quarter-wavelength transmitting monopole antenna being impedance matched at this frequency.

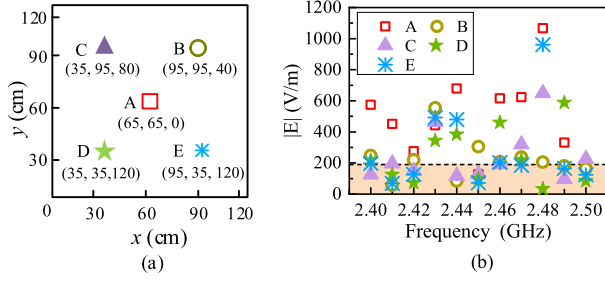


Fig. 3. (a) Positions of five targets within the ELC and (b) corresponding E-field magnitudes at those positions.

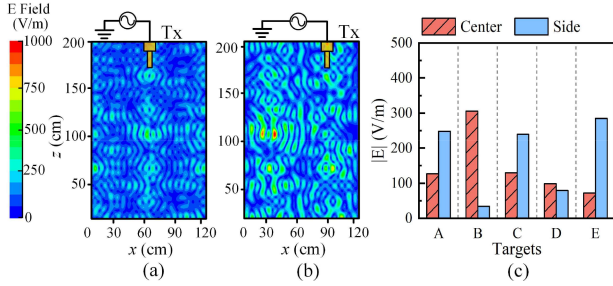


Fig. 4. E-field distributions at 2.45 GHz for transmitting antenna located at (a) top center and (b) offset position, and (c) comparison between the two configurations.

Fig. 3 compares the E-field magnitudes at five targets' locations (A–E) positioned at different heights. Each target exhibits a frequency-dependent field maximum, e.g., 2.48 GHz for target A, 2.43 GHz for target B, and 2.49 GHz for target D. These results confirm the feasibility of frequency-tuning MWPT tailored to multiple targets in ELCs.

A key advantage of the ELC is its ability to generate high EM field magnitudes without the path loss in free space propagation. This feature allows for multiple frequency options suitable for MWPT to the same target. As shown in Fig. 3(b), assuming the required E-field strength of 190 V/m for MWPT (corresponding to the desired power level), target A satisfies this condition at nine frequency points within the 2.4 GHz to 2.5 GHz range, while target B at six points, and target C at four points. This frequency diversity enables flexible selection of operating frequencies for optimal field strength at each target location.

To verify the sensitivity of the field distribution to the excitation position, Fig. 4(c) compares the E-field distributions for two positions: the top center (65 cm, 65 cm, 200 cm) and an offset location (105 cm, 65 cm, 200 cm). The results show that shifting the excitation position significantly alters the E-field pattern. For example, target B achieves a peak E-field of 300 V/m when the excitation is centered, while target E peaks at 285 V/m when the excitation is offset. These findings suggest that introducing multiple antennas with switchable feeding could offer further flexibility for MWPT field tailoring.

III. EXPERIMENTAL VERIFICATION

To experimentally validate the proposed method, a rectangular metal mesh ELC with dimensions of 1.3 m × 1.3 m × 2 m

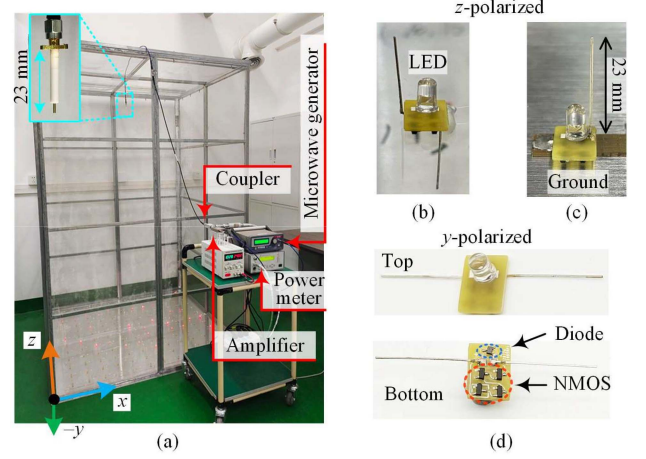


Fig. 5. Frequency-controlled MWPT inside the metal mesh ELC: (a) experimental setup, (b) z-polarized dipole, (c) z-polarized monopole, and (d) y-polarized dipole.

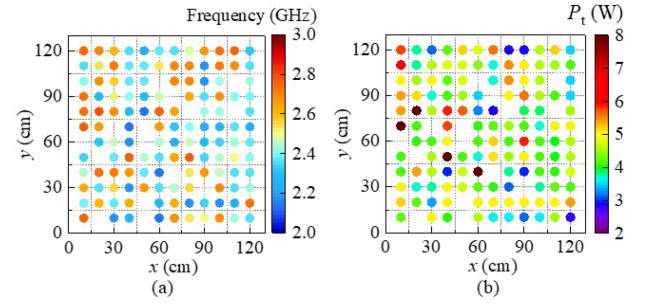


Fig. 6. Illumination of 130 y-directed LED rectennas by tuning (a) frequency and (b) transmitting power (P_t).

was constructed, as shown in Fig. 5(a). The metal mesh design is cost-effective and enables efficient MWPT monitoring for multiple targets. Its EM leakage is controlled to be less than 2 V/m for a maximum P_t of 1 W by limiting the grid spacing to less than 2.6 mm at 2.45 GHz [30]. Compared to PEC walls, the mesh reduces internal reflection, resulting in a moderately lower field magnitude in the ELC. The transmitting antenna is a 23 mm monopole, suspended 20 cm below the ELC ceiling.

LED rectennas are distributed as targets throughout the ELC, each comprising a Schottky diode (HSMS-282) and utilizing the LED's metal leads as receiving antennas. Three types were designed: z-polarized dipoles, z-polarized monopoles [Fig. 5(b) and (c)], and y-polarized dipoles [Fig. 5(d)]. For wall-mounted applications, the monopole design uses one lead bent upward and the other soldered to the ground. The measured rectification efficiency reaches approximately 40% within 2.4 GHz to 2.5 GHz when the rectenna receives an input microwave power of 20 mW. To prevent overvoltage, four NMOS transistors (model 2N7002) are integrated.

To facilitate MWPT observation, a 12 × 12 y-polarized LED rectenna array was placed at $z = 15$ mm [Fig. 5(a)]. As shown in Fig. 6, by adjusting the frequency and P_t , 130 out of 144 LEDs were selectively activated. Fig. 6(a) and (b) indicates that most LEDs can be lit up within the 2.4 GHz to 2.5 GHz band with

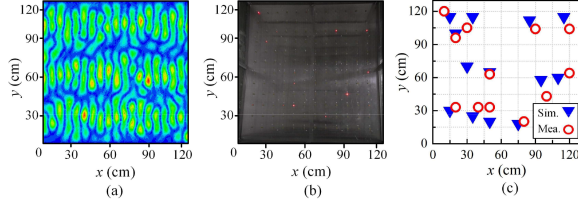


Fig. 7. Comparison between simulated and measured results at 2.4616 GHz: (a) simulated E_y component distribution at the $z = 15$ mm plane, (b) illuminated LEDs, and (c) positional correlation between high E_y spots and illuminated LEDs.

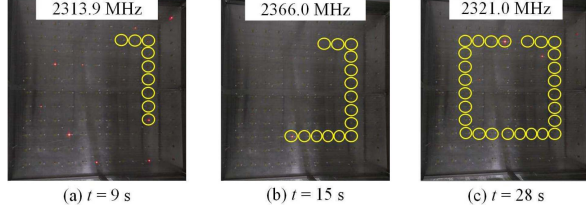


Fig. 8. Experimental results of frequency-tuned MWPT targeting multiple LEDs arranged along a rectangular path at (a) 9 s, (b) 15 s, and (c) 28 s.

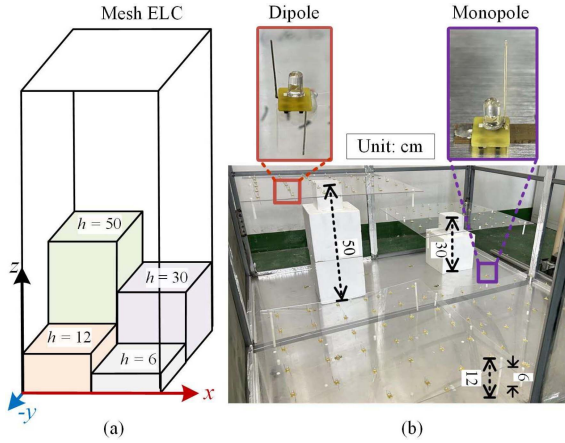


Fig. 9. Arrangement of 160 z -polarized targets at various heights: (a) schematic diagram and (b) photograph.

P_t of 4 W to 5 W. The high required P_t is due to polarization mismatch between the transmitter and rectennas.

Fig. 7 compares the simulated and measured results at 2.4616 GHz. The rectenna array is located on the x - y plane at $z = 15$ mm. Since the array is y -polarized, the simulated y -component of the E-field (E_y) is used for comparison. Good agreement is observed. Discrepancies are primarily attributed to the physical presence of 144 LED rectennas and to slight wall deformation of the metal mesh ELC in the experiment.

To further demonstrate the flexibility of the proposed method, an experiment was conducted with 29 LED rectennas arranged along a rectangular path. By sequentially applying optimal frequency and P_t settings, each target was illuminated for 1 second (partial results shown in Fig. 8). The sequential activation of targets, highlighted by yellow circles, validates the frequency-controlled multitarget MWPT.

Additionally, 160 LED rectennas were placed at five different heights in the ELC (0 cm, 6 cm, 12 cm, 30 cm, and 50 cm), as illustrated in Fig. 9. A 4×4 array of z -polarized monopole

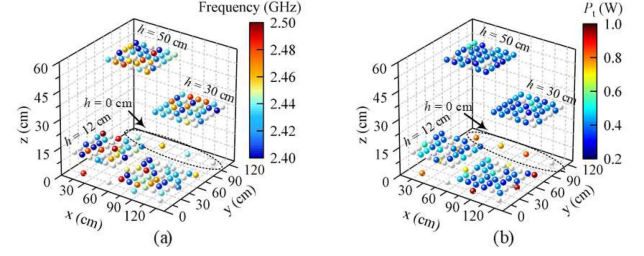


Fig. 10. Required (a) frequency and (b) power for the LEDs illumination.

TABLE I
PERFORMANCE COMPARISON

Ref.	Freq. (GHz)	Volume (λ^3)	Number of Targets	Tailored MWPT for Multiple Targets
[22]-2019	0.43	2.9	1	N. A
[29]-2020	0.45	0.6	4	N. A
[31]-2022	2.45	4.36	1	N. A
[33]-2014	10.0	0.9×10^3	1	Moderately Flexible
This work	2 – 3	1.8×10^3	160	Yes

λ is the wavelength at the resonant frequency in each Ref.
N. A. = Not applicable.

rectennas was placed at ground level, while four 6×6 arrays of z -polarized dipole LED rectennas were mounted on 2 mm-thick acrylic plates ($\epsilon_r = 2.67$) at higher elevations. As shown in Fig. 10(a) and (b), 111 out of 160 LEDs were illuminated using frequencies within 2.4 GHz to 2.5 GHz and P_t below 1 W. Unlit LEDs are marked with white spheres. Fig. 10(b) indicates that targets at $h = 0$ cm require higher P_t (> 700 mW) compared to suspended targets (400 mW to 600 mW). Table I compares the proposed method to related studies in publications. While most previous work involves fewer than four receivers, our system enables selective control of up to 160 targets in the largest ELC volume reported to date. The frequency-tunable MWPT strategy allows individualized power delivery to targets with varying energy demands.

V. CONCLUSION

This work presents a novel frequency-controlled MWPT method for tailored multitarget charging within an ELC of volume $1.8 \times 10^3 \times \lambda^3$. By leveraging the frequency sensitivity of the EM field distribution, the method enables tailored power delivery to multiple targets. Full-wave simulations confirm the effectiveness of frequency tuning in optimizing field strength at desired locations. Experimental results using a cost-effective metal mesh ELC and 160 LED rectennas show alignment with simulations. Sequential activation of 29 LEDs arranged along a rectangular path further validates the effective frequency-controlled MWPT. Unlike conventional power-based control, this frequency-based strategy dynamically shapes the MWPT path, ensuring efficient energy distribution while minimizing waste. The proposed MWPT technique is well suited for dense sensor networks in workplaces, such as reusable rockets, aircraft interiors, and industrial environments, offering a scalable and flexible solution for practical MWPT deployment.

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